

Synergism from sequenced combinations of curcumin and epigallocatechin-3-gallate with cisplatin in the killing of human ovarian cancer cells.

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Drug resistance remains an on-going challenge in ovarian cancer chemotherapy. The objective of this study was to determine the effect on synergism in activity from the sequenced combinations of cisplatin (Cis) with curcumin (Cur) and epigallocatechin-3-gallate (EGCG) in the human ovarian cancer cell lines. The drugs were added in binary combinations: Cis combined with Cur, and Cis combined with EGCG to the human ovarian A2780 and A2780(cisR) cancer cell lines, using five different sequences of administration: 0/0 h, 4/0 h, 0/4 h, 24/0 h and 0/24 h. The combination index (CI) was used to assess the combined action of the drugs. CIs <1, =1 and >1 indicated synergism, additiveness and antagonism respectively. Cellular accumulation of platinum and platinum-DNA binding levels from Cis and its combination with the phytochemicals were determined using graphite furnace atomic absorption spectrometry. Addition of Cis 4 h before Cur and EGCG (0/4 h combination) produced the most synergistic outcomes in both the A2780 and A2780(cisR) cell lines. The cellular accumulations of platinum and platinum-DNA binding resulting from the 0/4 h combinations were greater as compared to the values using Cis alone, thus providing an explanation for the synergistic action. When sequenced combinations of Cis with Cur and with EGCG are applied to human ovarian A2780 and A2780(cisR) cancer cell lines, lower concentrations and shorter time gap between the two additions seem to produce a higher cytotoxic effect.